Queue-Management-System-v2 — Pipeline B

YOLOv9 body detector + RetinaFace/Uniface face analytics + Norfair tracking with TimescaleDB logging.
What it does

- Detects people (bodies) in a configurable ROI polygon using YOLOv9 ONNX
- Detects handbags and backpacks (COCO classes 24 + 26) in the same pass
- Runs face attribute analysis (gender, age, confidence) via Uniface/RetinaFace
- Tracks individuals with Norfair (IoU-based)
- Per track, accumulates **confidence-weighted** gender votes and age readings
- Associates bags to persons via proximity check; `has_bag` is latched True once detected
- **Inserts one DB row per track at track death**
  - `timestamp` = actual entry time (first frame), not insert time
  - `gender` = confidence-weighted vote across all frames
  - `age_estimate` = confidence-weighted average across all frames
  - `has_bag` = True if a bag was detected near the person in any frame
  - Tracks shorter than `min_elapsed_time` are discarded without a DB insert
- Writes periodic `queue_state_snapshots` (queue count, avg/max dwell, lanes)

Setup

Download ONNX models and place them in `models/`:

https://drive.google.com/drive/folders/1gxqqcMACrjvegS0_OQypeT_lDKarco5_?usp=sharing

```
cd Queue-Management-System-v2-main/Queue-Management-System-v2-main
python -m venv venv
source venv/bin/activate          # Windows: venv\Scripts\activate
pip install -r requirements.txt
```

Run

```
python queue_management_v2.py
python queue_management_v2.py --execution_provider cuda
python queue_management_v2.py --source path/to/video.mp4 --view-img
```

Configuration files

File	Key settings
`config.yml`	`camID, ip_address, RTSP credentials, ONNX model path, ROI polygon, active_lanes`
`config2.yml`	`max_age, min_iou (face-body match), snapshot_interval, min_elapsed_time, debug_mode`

Camera ID note

Use `camID: SIM_live_test_video` when running against a local test video file so entries are excluded from real-data training queries.

Forecasting

```
python prophet_predict.py --source REAL
python ensemble_predict.py --source REAL
```

Database tables written

Table	Written when
`entrance_events`	At track death (one row per confirmed person, includes `has_bag`)
`queue_state_snapshots`	Every `snapshot_interval` seconds
`queue_predictions`	By `ensemble_predict.py` when run